

By defn. of conditional, $(p \vee q) \wedge \neg(p \wedge q) \rightarrow F$ is True.

$\therefore (p \vee q) \rightarrow (p \wedge q)$ is True

So, if both xy and $(x+y)$ are even, then both x and y are even. $\square$

Common Errors with Exhaustive Proof and Proof by Cases

A common error of reasoning is to draw incorrect conclusions from examples. No matter how many separate examples are considered, a theorem is not proved by considering examples unless every possible case is covered. The problem of proving a theorem is analogous to showing that a computer program always produces the output desired. No matter how many input values are tested, unless all input values are tested, we cannot conclude that the program always produces the correct output.

$\hookrightarrow V.O.Imp$

Example 8: Is it true that every positive integer is the sum of 18 fourth powers of integers?

Solution: To determine whether a positive integer n can be written as the sum of 18 fourth powers of integers, we might begin by examining whether n is the sum of 18 fourth powers of integers for the smallest positive integers. Because the fourth powers of integers are $0, 1, 16, 81, \dots$, if we can select 18 terms from these numbers that add up to n , then n is the sum of 18 fourth powers.

Claim: All positive integers up to 78 can be written as the sum of 18 fourth powers.

Proof:

$$\begin{aligned} 1 &= 1 + 0 + 0 + \dots + 0 \\ 2 &= 1 + 1 + 0 + \dots + 0 \\ 3 &= 1 + 1 + 1 + 0 + \dots + 0 \\ 4 &= 1 + 1 + 1 + 1 + 0 + \dots + 0 \\ 5 &= 5 \text{ 1's, } 13 \text{ 0's} \\ 6 &= 6 \text{ 1's, } 12 \text{ 0's} \\ 7 &= 7 \text{ 1's, } 11 \text{ 0's} \\ 8 &= 8 \text{ 1's, } 10 \text{ 0's} \\ 9 &= 9 \text{ 1's, } 9 \text{ 0's} \\ 10 &= 10 \text{ 1's, } 8 \text{ 0's} \\ 11 &= 11 \text{ 1's, } 7 \text{ 0's} \\ 12 &= 12 \text{ 1's, } 6 \text{ 0's} \\ 13 &= 13 \text{ 1's, } 5 \text{ 0's} \\ 14 &= 14 \text{ 1's, } 4 \text{ 0's} \end{aligned}$$